VIRTUALIZED NETWORK ENVIRONMENT USING MULTIPLE SERVICES

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Part A – Initial Setup

Setting up the required operating systems.
Server – CentOS, Client 1 – Ubuntu , Client 2 – Ubuntu

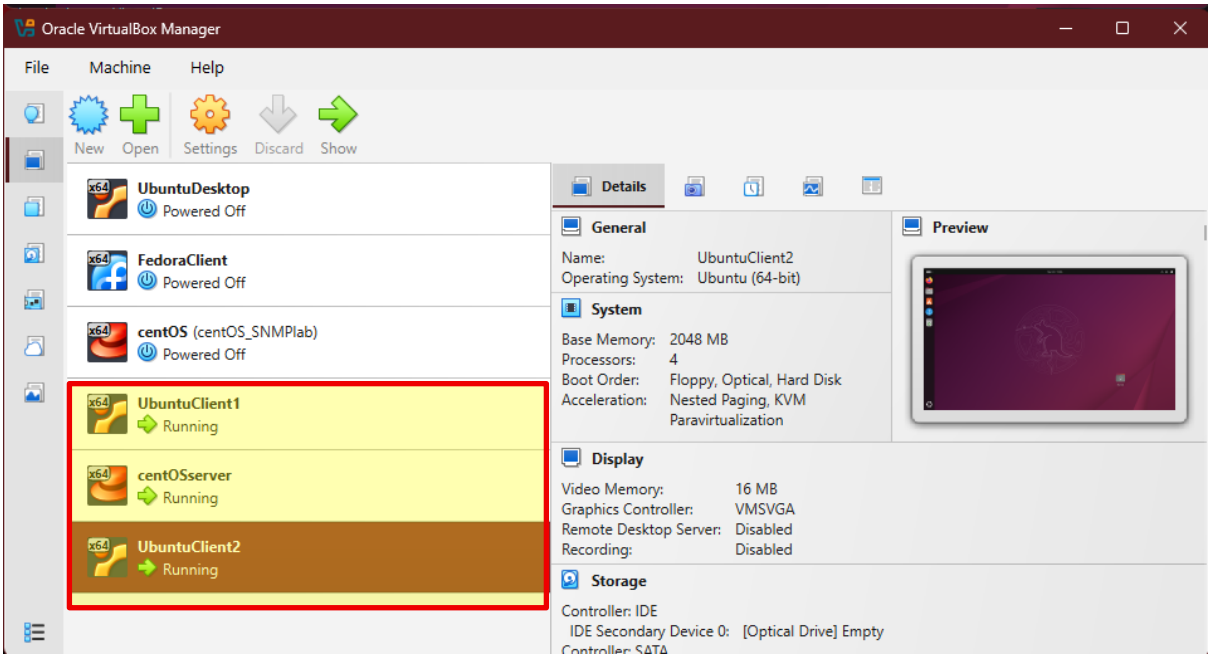

Figure 1.1: VirtualBox setup

Configuring network connectivity between the machines.

To establish connectivity between the machines, the network adaptor 1 is set to “internal Network” mode on all 3 VM’s

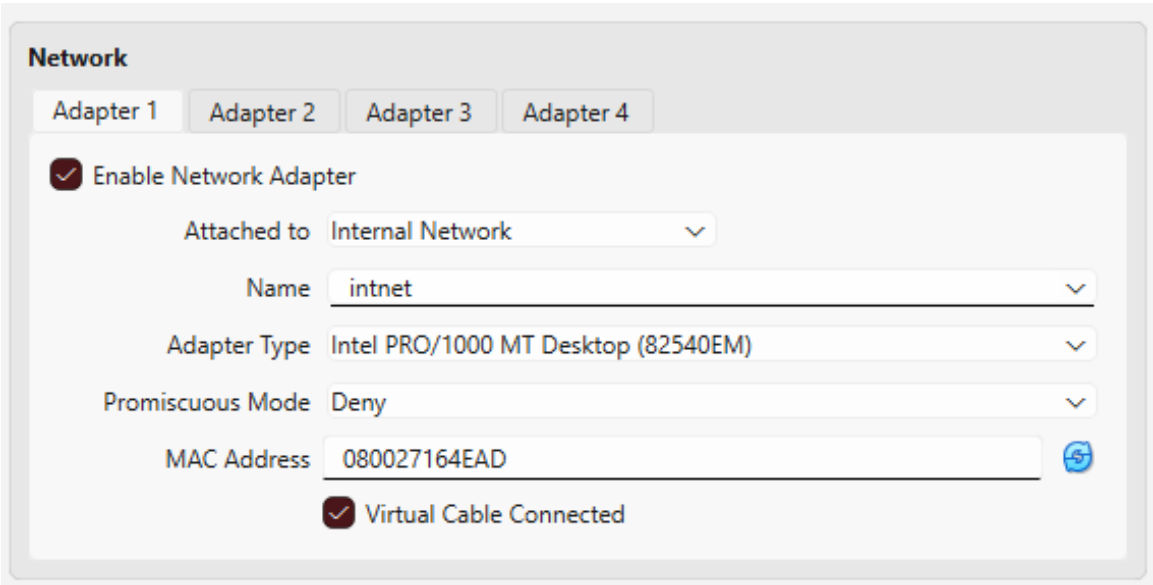

Figure 1.2: all 3 VM’s network setup

Configure temporary IP addresses to verify the connectivity between virtual machines
CentOS – 192.168.54.10
UbuntuClient1 – 192.168.54.30
UbuntuClient2 – 192.168.54.31

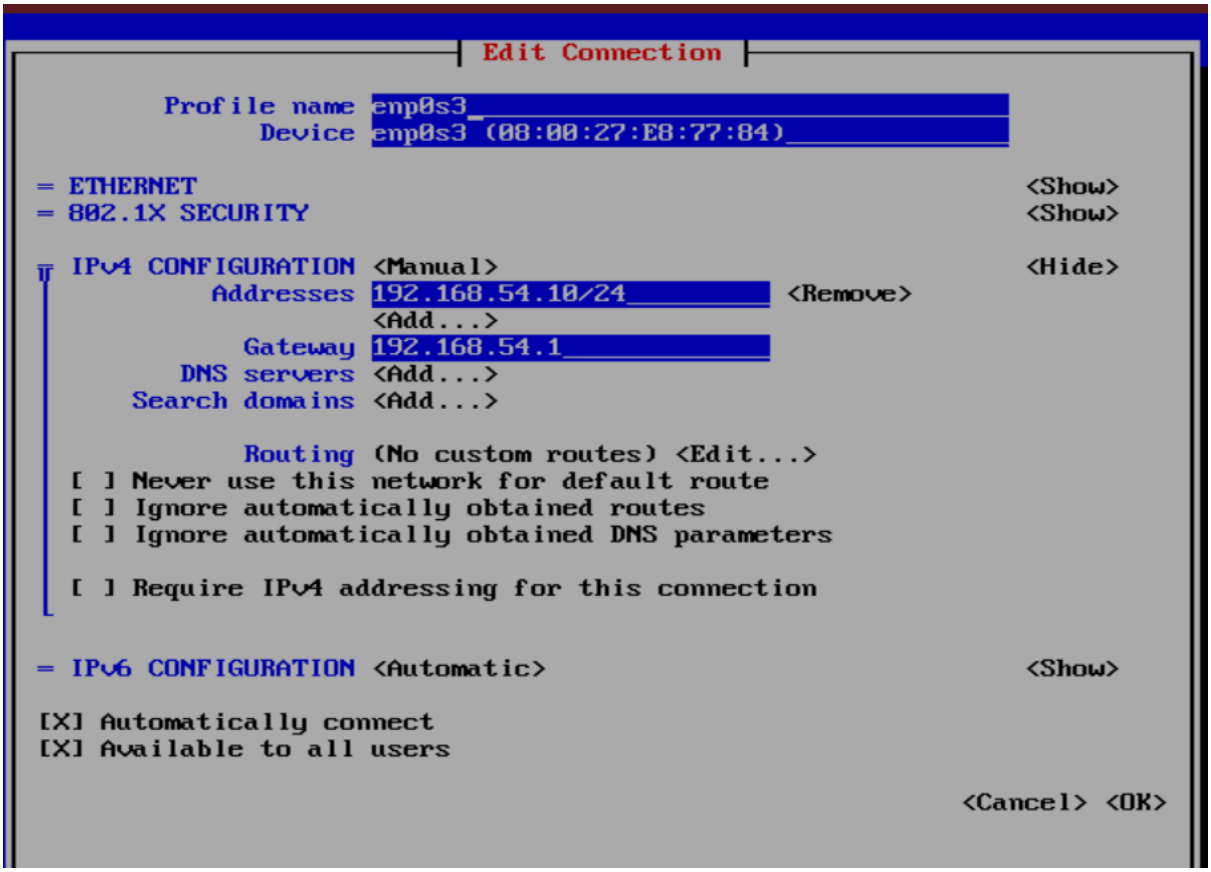

Figure 1.3: CentOS ip interfaces

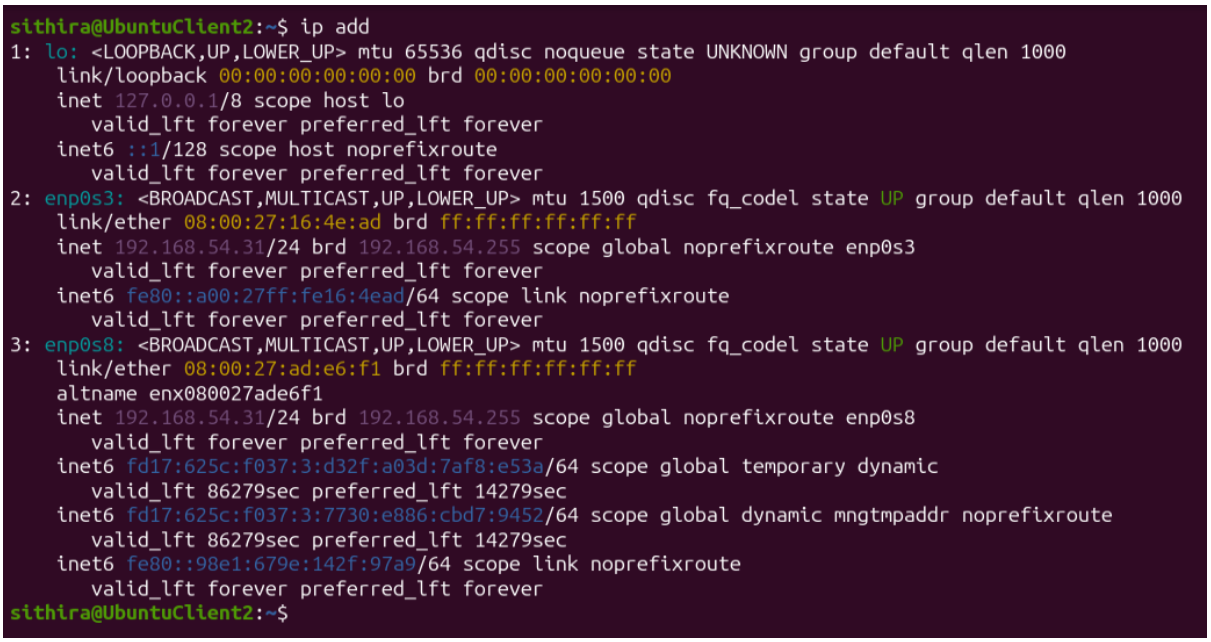

Figure 1.4: Client 2 ip interfaces

Verify network connectivity using the ping command.

```
sithira@UbuntuClient2:~$ ping 192.168.54.10
PING 192.168.54.10 (192.168.54.10) 56(84) bytes of data.
64 bytes from 192.168.54.10: icmp_seq=1 ttl=64 time=2.99 ms
64 bytes from 192.168.54.10: icmp_seq=2 ttl=64 time=1.38 ms
64 bytes from 192.168.54.10: icmp_seq=3 ttl=64 time=1.63 ms
64 bytes from 192.168.54.10: icmp_seq=4 ttl=64 time=3.61 ms
64 bytes from 192.168.54.10: icmp_seq=5 ttl=64 time=2.04 ms
64 bytes from 192.168.54.10: icmp_seq=6 ttl=64 time=0.955 ms
64 bytes from 192.168.54.10: icmp_seq=7 ttl=64 time=0.768 ms
64 bytes from 192.168.54.10: icmp_seq=8 ttl=64 time=1.66 ms
64 bytes from 192.168.54.10: icmp_seq=9 ttl=64 time=1.11 ms
64 bytes from 192.168.54.10: icmp_seq=10 ttl=64 time=1.10 ms
64 bytes from 192.168.54.10: icmp_seq=11 ttl=64 time=1.13 ms
64 bytes from 192.168.54.10: icmp_seq=12 ttl=64 time=1.10 ms
64 bytes from 192.168.54.10: icmp_seq=13 ttl=64 time=3.44 ms
64 bytes from 192.168.54.10: icmp_seq=14 ttl=64 time=1.15 ms
64 bytes from 192.168.54.10: icmp_seq=15 ttl=64 time=2.37 ms
64 bytes from 192.168.54.10: icmp_seq=16 ttl=64 time=0.757 ms
64 bytes from 192.168.54.10: icmp_seq=17 ttl=64 time=1.57 ms
64 bytes from 192.168.54.10: icmp_seq=18 ttl=64 time=2.96 ms
^C
--- 192.168.54.10 ping statistics ---
18 packets transmitted, 18 received, 0% packet loss, time 17123ms
rtt min/avg/max/mdev = 0.757/1.762/3.613/0.899 ms
sithira@UbuntuClient2:~$
```

```
sithira@UbuntuClient1:~$ ping 192.168.54.10
PING 192.168.54.10 (192.168.54.10) 56(84) bytes of data.
64 bytes from 192.168.54.10: icmp_seq=1 ttl=64 time=1.35 ms
64 bytes from 192.168.54.10: icmp_seq=2 ttl=64 time=3.80 ms
64 bytes from 192.168.54.10: icmp_seq=3 ttl=64 time=2.73 ms
64 bytes from 192.168.54.10: icmp_seq=4 ttl=64 time=1.31 ms
64 bytes from 192.168.54.10: icmp_seq=5 ttl=64 time=1.59 ms
64 bytes from 192.168.54.10: icmp_seq=6 ttl=64 time=4.44 ms
64 bytes from 192.168.54.10: icmp_seq=7 ttl=64 time=1.88 ms
64 bytes from 192.168.54.10: icmp_seq=8 ttl=64 time=1.59 ms
64 bytes from 192.168.54.10: icmp_seq=9 ttl=64 time=1.91 ms
64 bytes from 192.168.54.10: icmp_seq=10 ttl=64 time=1.47 ms
64 bytes from 192.168.54.10: icmp_seq=11 ttl=64 time=2.35 ms
64 bytes from 192.168.54.10: icmp_seq=12 ttl=64 time=2.28 ms
64 bytes from 192.168.54.10: icmp_seq=13 ttl=64 time=2.29 ms
64 bytes from 192.168.54.10: icmp_seq=14 ttl=64 time=2.05 ms
64 bytes from 192.168.54.10: icmp_seq=15 ttl=64 time=1.60 ms
64 bytes from 192.168.54.10: icmp_seq=16 ttl=64 time=1.82 ms
64 bytes from 192.168.54.10: icmp_seq=17 ttl=64 time=1.77 ms
64 bytes from 192.168.54.10: icmp_seq=18 ttl=64 time=2.53 ms
^C
--- 192.168.54.10 ping statistics ---
18 packets transmitted, 18 received, 0% packet loss, time 22331ms
rtt min/avg/max/mdev = 1.305/2.152/4.435/0.805 ms
sithira@UbuntuClient1:~$
```

Figure 1.5: Client 1 and 2 successfully ping Server's ip address (192.168.54.10)

```
centOSserver [Running] - Oracle VirtualBox
[root@localhost ~]# ping 192.168.54.31
PING 192.168.54.31 (192.168.54.31) 56(84) bytes of data.
64 bytes from 192.168.54.31: icmp_seq=1 ttl=64 time=5.69 ms
64 bytes from 192.168.54.31: icmp_seq=2 ttl=64 time=1.61 ms
64 bytes from 192.168.54.31: icmp_seq=3 ttl=64 time=1.65 ms
64 bytes from 192.168.54.31: icmp_seq=4 ttl=64 time=2.24 ms
^C
--- 192.168.54.31 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3266ms
rtt min/avg/max/mdev = 1.609/2.795/5.691/1.690 ms
[root@localhost ~]# ping 192.168.54.30
PING 192.168.54.30 (192.168.54.30) 56(84) bytes of data.
64 bytes from 192.168.54.30: icmp_seq=1 ttl=64 time=5.75 ms
64 bytes from 192.168.54.30: icmp_seq=2 ttl=64 time=5.41 ms
64 bytes from 192.168.54.30: icmp_seq=3 ttl=64 time=2.10 ms
64 bytes from 192.168.54.30: icmp_seq=4 ttl=64 time=2.58 ms
64 bytes from 192.168.54.30: icmp_seq=5 ttl=64 time=2.35 ms
64 bytes from 192.168.54.30: icmp_seq=6 ttl=64 time=0.945 ms
^C
--- 192.168.54.30 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5013ms
rtt min/avg/max/mdev = 0.945/3.187/5.749/1.769 ms
[root@localhost ~]# _
```

Figure 1.6: CentOS successfully pings both client IP's

Part B – DHCP Server

As the DHCP server, Kea DHCP server was used.

centOSserver [Running] - Oracle VirtualBox

```
{
  "Dhcp4": {
        "interfaces-config": {
                "interfaces": [ "enp0s3" ]
        },

        "lease-database": {
          "type": "memfile",
          "persist": true,
          "name": "/var/lib/kea/kea-leases4.csv"
        },

        "valid-lifetime": 600,

        "subnet4": [
          {
              "id": 1,
              "subnet": "192.168.54.0/24",

              "pools": [
                {
                  "pool": "192.168.54.100 - 192.168.54.200"
                }
              ],

              "option-data": [
               {
                 "name": "routers",
                 "data": "192.168.54.10"
               },
               {
                 "name": "subnet-mask",
                 "data": "255.255.255.0"
               },
               {
                 "name": "domain-name-servers",
                 "data": "8.8.8.8"
               }
              ]
          }
        ]
      }
}
~
~
```

Figure 2.1: Kea DHCP configuration file (/etc/kea/kea-dhcp4.conf)

```
~
~
● kea-dhcp4.service - Kea DHCPv4 Server
     Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-03-29 13:36:23 +0530; 11min ago
 Invocation: 465f617428714cd492e9852a5bd9ab1f
       Docs: man:kea-dhcp4(8)
   Main PID: 1979 (kea-dhcp4)
     Status: "Dispatching packets..."
      Tasks: 7 (limit: 22990)
     Memory: 2.8M (peak: 6M)
        CPU: 227ms
     CGroup: /system.slice/kea-dhcp4.service
             └─1979 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

Mar 29 13:40:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:40:39.348 INFO  [kea-dhcp4.leases/1979.140224310687424] DHCP4_LEASE_OFFER [hwtype=1 08:00:
Mar 29 13:40:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:40:39.349 INFO  [kea-dhcp4.packets/1979.140224310687424] DHCP4_PACKET_SEND [hwtype=1 08:00
Mar 29 13:40:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:40:39.352 INFO  [kea-dhcp4.dhcp4/1979.140224319080128] DHCP4_QUERY_LABEL received query: [
Mar 29 13:40:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:40:39.352 INFO  [kea-dhcp4.packets/1979.140224319080128] DHCP4_PACKET_RECEIVED [hwtype=1 0
Mar 29 13:40:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:40:39.354 INFO  [kea-dhcp4.leases/1979.140224319080128] DHCP4_LEASE_ALLOC [hwtype=1 08:00:
Mar 29 13:40:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:40:39.354 INFO  [kea-dhcp4.packets/1979.140224319080128] DHCP4_PACKET_SEND [hwtype=1 08:00
Mar 29 13:45:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:45:39.401 INFO  [kea-dhcp4.dhcp4/1979.140224310687424] DHCP4_QUERY_LABEL received query: [
Mar 29 13:45:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:45:39.410 INFO  [kea-dhcp4.packets/1979.140224310687424] DHCP4_PACKET_RECEIVED [hwtype=1 0
Mar 29 13:45:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:45:39.427 INFO  [kea-dhcp4.leases/1979.140224310687424] DHCP4_LEASE_ALLOC [hwtype=1 08:00:
Mar 29 13:45:39 localhost.localdomain kea-dhcp4[1979]: 2026-03-29 13:45:39.430 INFO  [kea-dhcp4.packets/1979.140224310687424] DHCP4_PACKET_SEND [hwtype=1 08:00
~
lines 1-23/23 (END)
```

Figure 2.2: Kea DHCP service is running successfully on the CentOS server

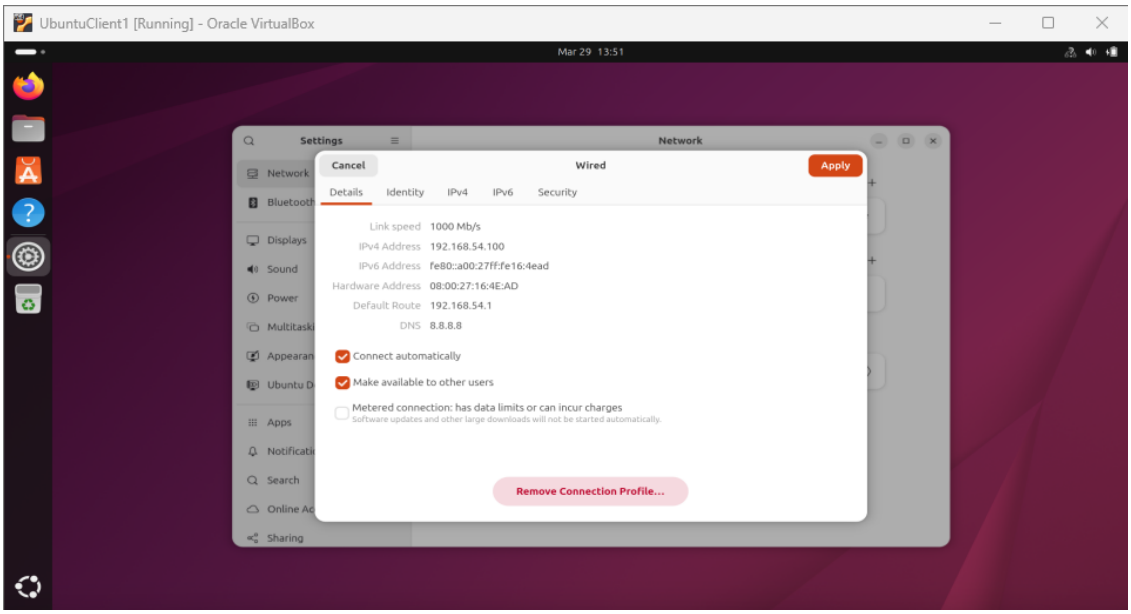

Figure 2.3: Ubuntu client 1 assigned with 192.168.54.100 ip address by DHCP

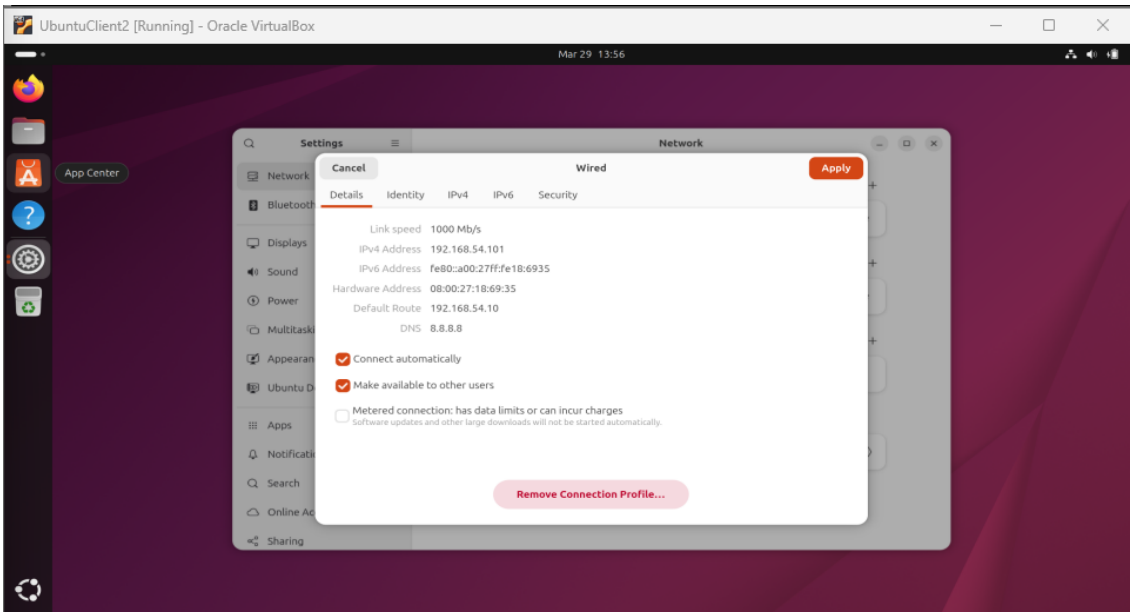

Figure 2.4: Ubuntu client 2 assigned with 192.168.54.101 ip address by DHCP

```
[root@localhost ~]# cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
192.168.54.100,08:00:27:16:4e:ad,01:08:00:27:16:4e:ad,600,1774772439,1,0,0,ubuntuclient1,0,,0
192.168.54.100,08:00:27:16:4e:ad,01:08:00:27:16:4e:ad,600,1774772739,1,0,0,ubuntuclient1,0,,0
192.168.54.100,08:00:27:16:4e:ad,01:08:00:27:16:4e:ad,600,1774773039,1,0,0,ubuntuclient1,0,,0
192.168.54.100,08:00:27:16:4e:ad,01:08:00:27:16:4e:ad,600,1774773137,1,0,0,ubuntuclient2,0,,0
192.168.54.101,08:00:27:18:69:35,01:08:00:27:18:69:35,600,1774773334,1,0,0,ubuntuclient2,0,,0
192.168.54.100,08:00:27:16:4e:ad,01:08:00:27:16:4e:ad,600,1774773137,1,0,0,,2,,0
[root@localhost ~]#
```

Figure 2.5: IP lease file (/var/lib/kea/kea-leases4.csv)

The client machines were configured to obtain an IP address automatically using DHCP. Upon connecting to the network, it successfully received an IP address within the defined range from the DHCP server, as verified through system commands and server lease records.

PART C - Installing zimbra

A dedicated Ubuntu 20.04.6 server was deployed as a virtual machine in Oracle VirtualBox to host the Zimbra mail server.

```
sithira@ubuntu20server:~$ sudo hostnamectl set-hostname mail.it54.local
sithira@ubuntu20server:~$ hostname
mail.it54.local
sithira@ubuntu20server:~$
```

Figure 3.1: hostname

configuration of the mail server hostname as mail.it54.local using the hostnamectl command, before Zimbra installation.

```
  GNU nano 4.8                    /etc/hosts                    Modified
127.0.0.1 localhost
127.0.1.1 ubuntu20server
192.168.54.20 mail.it54.local.mail
# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

Figure 3.2: the /etc/hosts file configuration

mapping the server IP address to the domain name mail.it54.local to ensure proper DNS resolution.

```
sithira@ubuntu20server:~$ wget https://files.zimbra.com/downloads/10.0.0_GA/zcs-NETWORK-10.0.0_GA_45
18.UBUNTU20_64.20230301065514.tgz
--2026-04-05 19:52:44--  https://files.zimbra.com/downloads/10.0.0_GA/zcs-NETWORK-10.0.0_GA_4518.UBU
NTU20_64.20230301065514.tgz
Resolving files.zimbra.com (files.zimbra.com)... 99.86.16.151
Connecting to files.zimbra.com (files.zimbra.com)|99.86.16.151|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 467583331 (446M) [application/x-tar]
Saving to: 'zcs-NETWORK-10.0.0_GA_4518.UBUNTU20_64.20230301065514.tgz'

cs-NETWORK-10.0.0_GA_451  75%[=========================>         ] 334.58M  17.6MB/s    eta 7s
```

Figure 3.3: downloading Zimbra

download of the Zimbra Collaboration Suite package using the wget command from the official Zimbra website.

```
  15) POP server port:                        7110
  16) POP server SSL port:                    7995
  17) Use spell check server:                 yes
  18) Spell server URL:                       http://mail.it54.local:7780/aspell.php
  19) Enable version update checks:           TRUE
  20) Enable version update notifications:    TRUE
  21) Version update notification email:      admin@it54.local
  22) Version update source email:            admin@it54.local
  23) Install mailstore (service webapp):     yes
  24) Install UI (zimbra,zimbraAdmin webapps): yes

Select, or 'r' for previous menu [r] r

Main menu

   1) Common Configuration:
   2) zimbra-ldap:                            Enabled
   3) zimbra-logger:                          Enabled
   4) zimbra-mta:                             Enabled
   5) zimbra-dnscache:                        Enabled
   6) zimbra-snmp:                            Enabled
   7) zimbra-store:                           Enabled
   8) zimbra-spell:                           Enabled
   9) zimbra-proxy:                           Enabled
  10) Default Class of Service Configuration:
   s) Save config to file
   x) Expand menu
   q) Quit

*** CONFIGURATION COMPLETE - press 'a' to apply
Select from menu, or press 'a' to apply config (? - help) a
Save configuration data to a file? [Yes] yes
Save config in file: [/opt/zimbra/config.23319]
Saving config in /opt/zimbra/config.23319...done.
The system will be modified - continue? [No] yes
Operations logged to /tmp/zmsetup.20260405-204839.log
Setting local config values...
```

Figure 3.4: Zimbra menu

Zimbra installation configuration menu where services such as LDAP, MTA, and mailbox are enabled before applying the configuration.

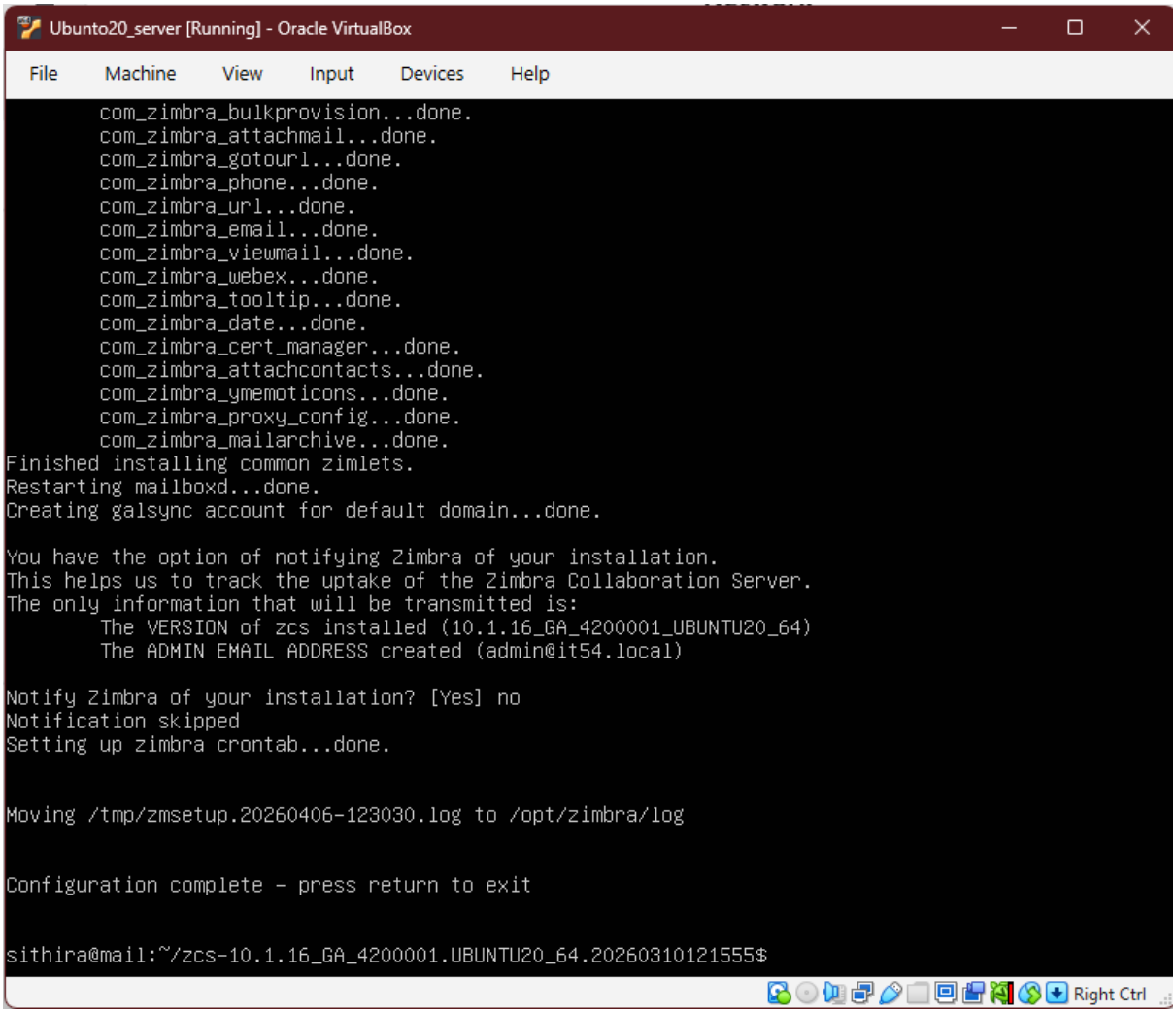

Figure 3.5: Zimbra installation

successful installation of Zimbra services and creation of the default admin account for the mail domain.

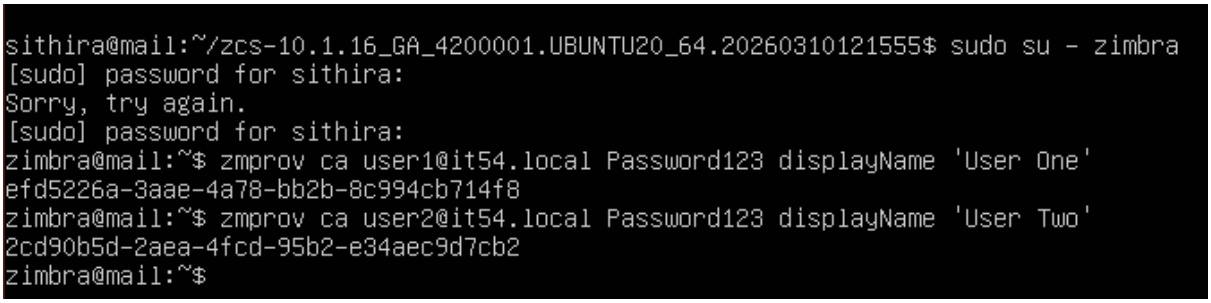

Figure 3.6: user creation

creation of user email accounts (user1@it54.local and user2@it54.local) using Zimbra command-line tools.

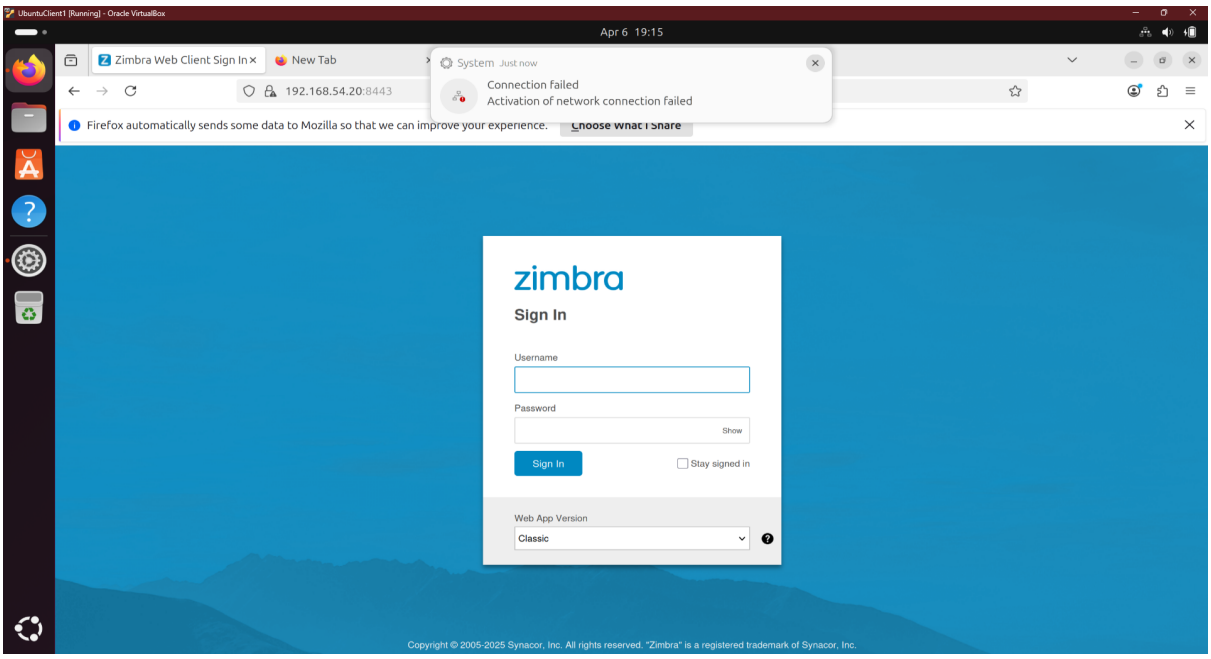

Figure 3.7: login interface

Zimbra webmail login interface accessed via a Ubuntu Client1's web browser using the server IP address.

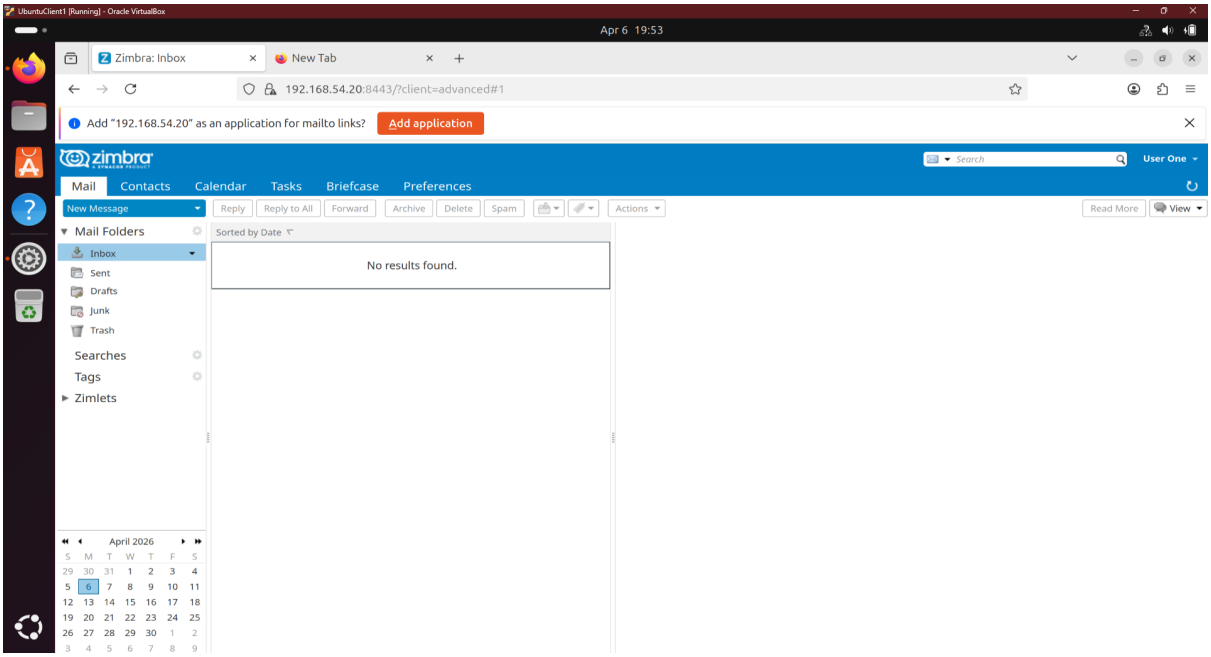

Figure 3.8: mailbox

Zimbra mailbox interface after successful login, showing the user dashboard and mail folders.

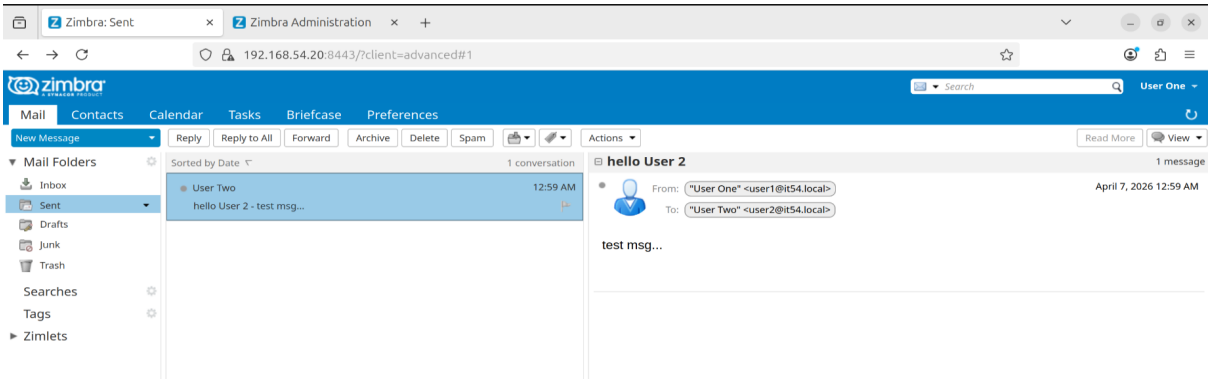

Figure 3.9: User One sending email

email sent from one user account to another within the Zimbra system, demonstrating email functionality.

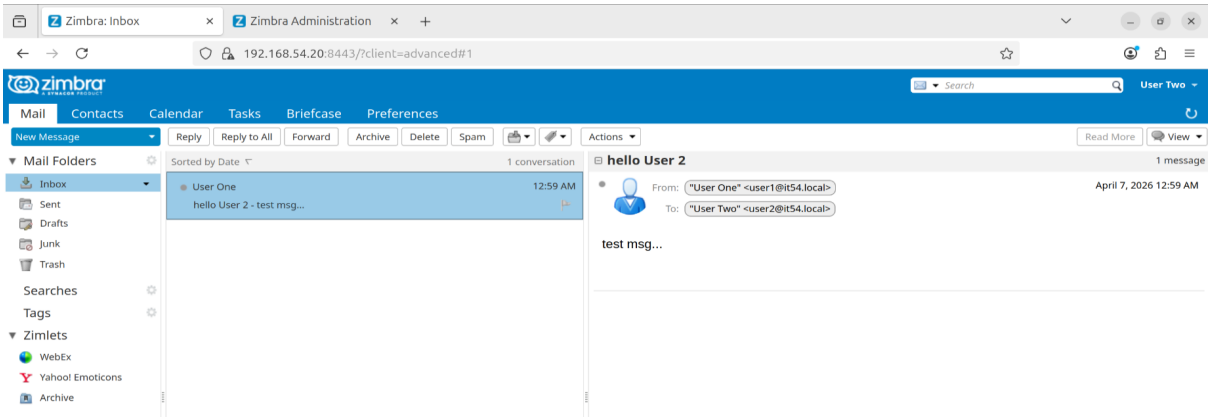

Figure 3.10: User two receiving email

received email in the recipient's inbox, confirming successful email communication between users.

PART D – SAMBA Active Directory Domain Controller

for the SAMBA server, a new Ubuntu24.04 server has been deployed

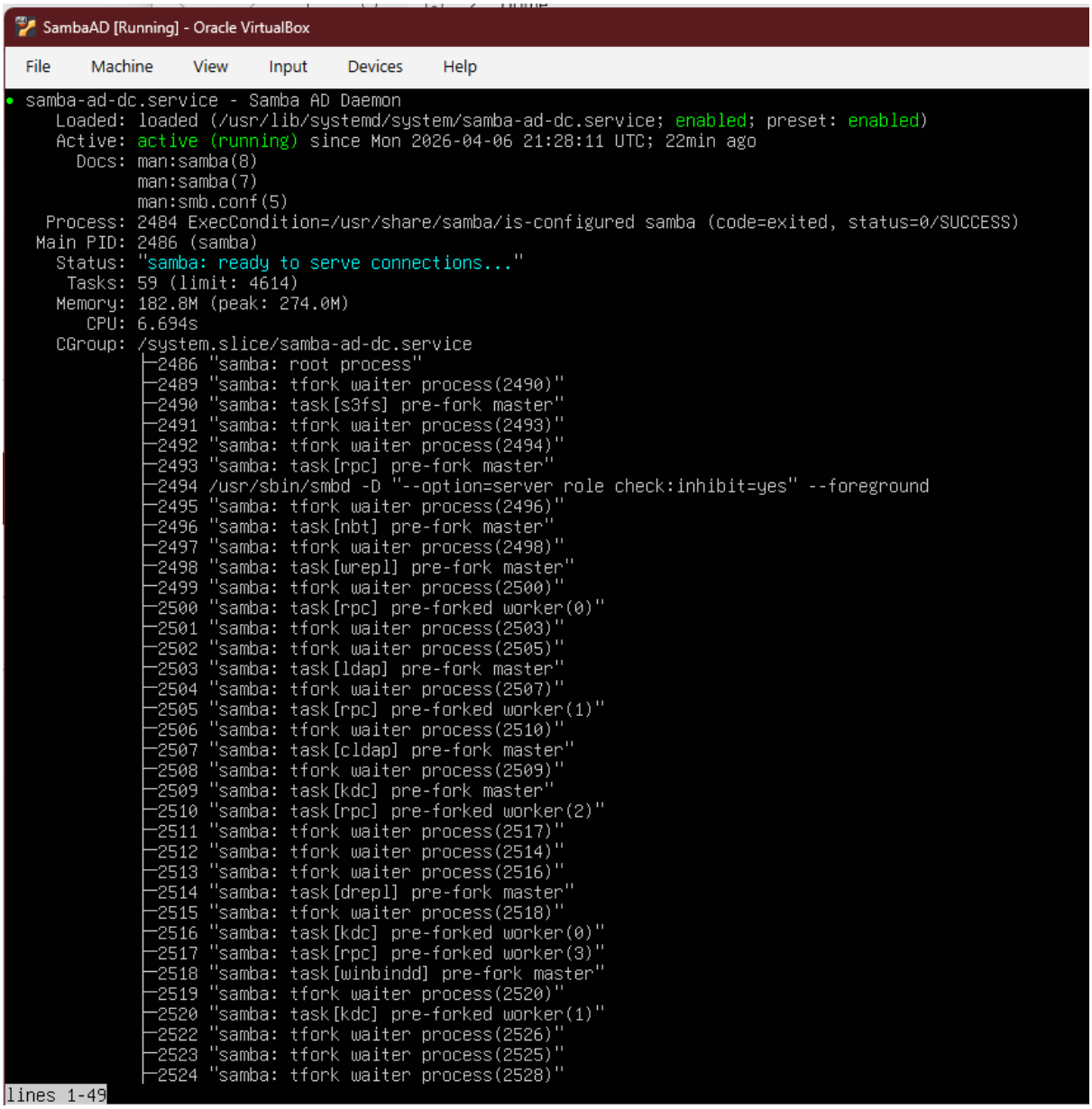

Figure 4.1: SAMBA status

After installing, Samba Active Directory Domain Controller service running successfully

```
sithira@SambaAD:~$ sudo samba-tool ou create "OU=Admins,DC=it54,DC=local"
Added ou "OU=Admins,DC=it54,DC=local"
sithira@SambaAD:~$ sudo samba-tool ou create "OU=Staff,DC=it54,DC=local"
Added ou "OU=Staff,DC=it54,DC=local"
sithira@SambaAD:~$ sudo samba-tool ou create "OU=Students,DC=it54,DC=local"
Added ou "OU=Students,DC=it54,DC=local"
sithira@SambaAD:~$ sudo samba-tool user create admin1 Password@123
User 'admin1' added successfully
sithira@SambaAD:~$ sudo samba-tool user create staff1 Password@123
User 'staff1' added successfully
sithira@SambaAD:~$ sudo samba-tool user create student1 Password@123
User 'student1' added successfully
sithira@SambaAD:~$ sudo samba-tool user move admin1 "OU=Admins,DC=it54,DC=local"
Moved user "admin1" into "OU=Admins,DC=it54,DC=local"
sithira@SambaAD:~$ sudo samba-tool user move staff1 "OU=Staff,DC=it54,DC=local"
Moved user "staff1" into "OU=Staff,DC=it54,DC=local"
sithira@SambaAD:~$ sudo samba-tool user move student1 "OU=Students,DC=it54,DC=local"
Moved user "student1" into "OU=Students,DC=it54,DC=local"
sithira@SambaAD:~$ _
```

Figure 4.2: creation of organizational units

creation of Organizational Units (Admins, Staff, Students) and user accounts (admin1, staff1, student1). It also shows users being assigned to their respective OUs using Samba commands.

```
sithira@UbuntuClient1:~$ host SambaAD.it54.local
SambaAD.it54.local has address 192.168.54.30
sithira@UbuntuClient1:~$ ping SambaAD.it54.local
PING SambaAD.it54.local (192.168.54.30) 56(84) bytes of data.
64 bytes from 192.168.54.30: icmp_seq=1 ttl=64 time=0.947 ms
64 bytes from 192.168.54.30: icmp_seq=2 ttl=64 time=0.855 ms
64 bytes from 192.168.54.30: icmp_seq=3 ttl=64 time=1.74 ms
64 bytes from 192.168.54.30: icmp_seq=4 ttl=64 time=1.59 ms
^C
--- SambaAD.it54.local ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 12025ms
rtt min/avg/max/mdev = 0.855/1.282/1.736/0.386 ms
```

Figure 4.3: connectivity test

successful DNS resolution and network connectivity between the client machine and the Samba AD server

```
sithira@UbuntuClient1:~$ realm list
it54.local
  type: kerberos
  realm-name: IT54.LOCAL
  domain-name: it54.local
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@it54.local
  login-policy: allow-realm-logins
sithira@UbuntuClient1:~$ S
```

Figure 4.4: client machine

client machine has successfully joined the IT54.LOCAL domain

```
sithira@UbuntuClient1:~$ klist
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: Administrator@IT54.LOCAL

Valid starting       Expires              Service principal
04/07/2026 04:37:30  04/07/2026 14:37:30  krbtgt/IT54.LOCAL@IT54.LOCAL
        renew until 04/08/2026 04:37:17
sithira@UbuntuClient1:~$
```

```
sithira@UbuntuClient1:~$ id Administrator@it54.local
uid=704400500(administrator@it54.local) gid=704400513(domain users@it54.local) g
roups=704400513(domain users@it54.local),704400520(group policy creator owners@i
t54.local),704400519(enterprise admins@it54.local),704400512(domain admins@it54.
local),704400572(denied rodc password replication group@it54.local),704400518(sc
hema admins@it54.local)
sithira@UbuntuClient1:~$
```

Figure 4.5: domain user account

details of the domain user account retrieved using the id command, confirming successful domain integration and user recognition.

Part E - Web Server

The web server is installed on the same server where Samba is installed.

```
sithira@SambaAD:~$ sudo systemctl start apache2
sithira@SambaAD:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
sithira@SambaAD:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-04-06 23:27:18 UTC; 1min 20s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3894 (apache2)
      Tasks: 55 (limit: 4614)
     Memory: 5.4M (peak: 5.7M)
        CPU: 72ms
     CGroup: /system.slice/apache2.service
             ├─3894 /usr/sbin/apache2 -k start
             ├─3896 /usr/sbin/apache2 -k start
             └─3897 /usr/sbin/apache2 -k start
sithira@SambaAD:~$
```

Figure 5.1: Apache server status

After Installing, Apache2 web server running on the Ubuntu server

```
  GNU nano 7.2                                                    /var/www/html/index.html
<html>
<head>
<title>HELLO NMA</title>
<head>
<body style="background-color:blue;">
<h1 style=color:white; text-align:center;; margin-top:20%;">
HELLO NMA
</h1>
</body>
</html>
```

Figure 5.2: index.html file

Creating the web page in index.html file

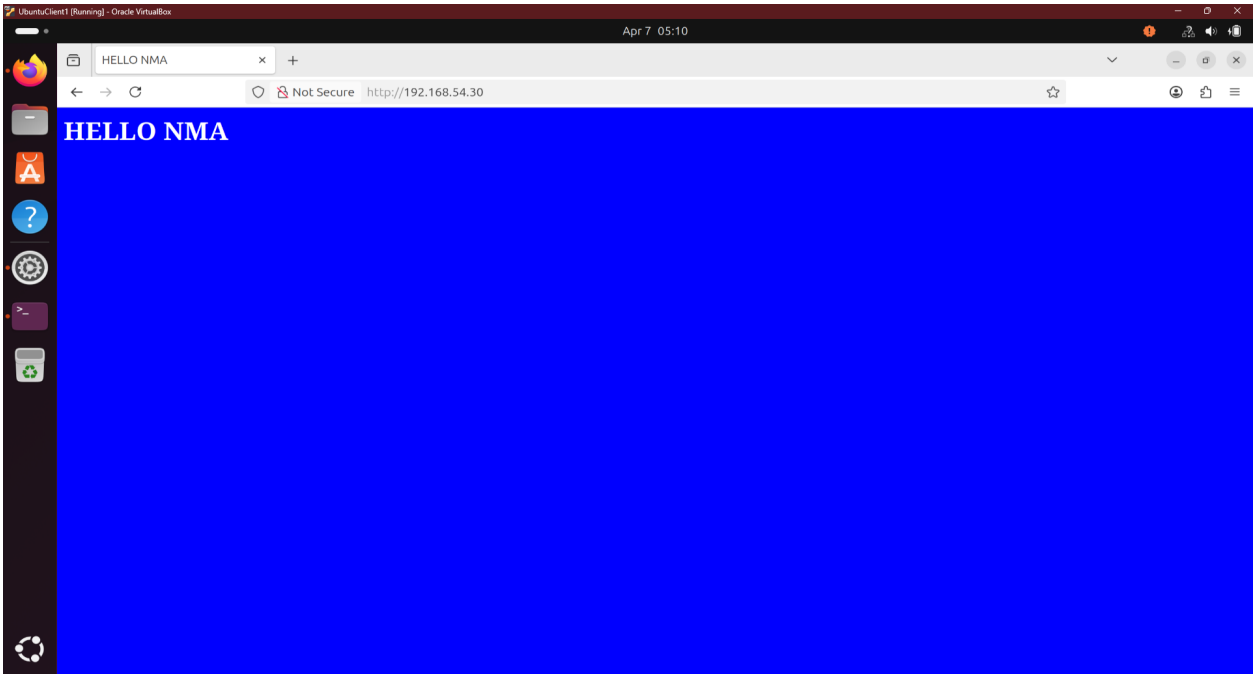

Figure 5.3: webpage

webpage is accessible from the client machine using a web browser. In here Ubuntu Client 1 machine is used. And access the web page via http://192.168.54.30 which is Samba/Apache web server ip address

Conclusion

In this assignment, a complete virtualized network environment was successfully designed and implemented using Oracle VirtualBox. Initially, multiple virtual machines were configured with appropriate operating systems, and internal network connectivity was established and verified using manual IP addressing and ping tests.

A Kea DHCP server was then configured on the CentOS server to dynamically assign IP addresses to client machines. This was tested and confirmed through successful IP allocation and lease file verification. Following that, a Zimbra mail server was deployed on a dedicated Ubuntu 20.04.6 virtual machine. The hostname and hosts file were properly configured to ensure correct domain name resolution, and Zimbra services were installed and initialized. User accounts were created, and email communication between users was successfully tested through the webmail interface.

In addition, a Samba Active Directory Domain Controller was set up on an Ubuntu 24.04 server. Organizational Units and user accounts were created, and a client machine was successfully joined to the domain, verifying proper authentication and directory services functionality.

Finally, an Apache web server was installed on the same server as the Samba service. A basic web page was created and hosted, and accessibility from a client machine was confirmed using a web browser.

Overall, this assignment provided practical experience in configuring and integrating multiple network services within a controlled virtual environment. It enhanced understanding of network administration concepts such as IP addressing, DHCP, DNS resolution, domain management, and service deployment.

References

1. Zimbra Official Documentation – https://www.zimbra.com
2. Kea DHCP Documentation – https://kea.readthedocs.io
3. Samba Documentation – https://www.samba.org
4. Apache HTTP Server Documentation – https://httpd.apache.org